

ARTIFICIAL INTELLIGENCE

Getting Real, Local



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Artificial intelligence (AI) is undoubtedly one of the hottest verticals to watch in 2017. An integral part of gaming and robotics, now it has also deeply penetrated aspects of the real world that influence everybody. As the core driving force of analytics, AI is at the heart of every vertical from industrial operations and banking to security and the network of devices. It has also become part and parcel of consumer electronics, as was evident from this year's CES 2017, and an inseparable part of social media as we can see from the new features being launched by Facebook, Google and other Internet majors. The increased usage of AI technologies fuels further growth in the space—not just in the form of research but also new products such as chips for artificial intelligence. The arena is also teeming with start-ups that are bringing fresh new ideas to life.

Some see AI as a threat to their jobs, while others see it as an opportunity to do better. Either way, it is not something you can ignore. So, well, we try to give you a glimpse of what is happening here...

AI shakes up the chip market

The word 'smart' has become like a compulsory prefix for most object names, from pens to phones and musical instruments.

The moment the word 'smart' is used, it means some amount of intelligence (obviously artificial) is involved. This intelligence is usually implemented using large servers with built-in AI models based on neural networks.

In general, the operation of neural networks can be understood in two stages. The first stage involves training the AI system to perform a given task. The second stage is the inference stage, which involves execution of AI models against live data.

The training stage involves crunching immensely large data sets, to help the system identify and learn from patterns in the data. Data collected by the device flies across the Internet to data centres packed with servers, where this is done. On servers, central processing units (CPUs) are usually supplemented with other types of processors like graphical processing units (GPUs).

Nvidia is one of the most well-known GPU makers, who has been powering many data centres for years. Last year, it launched Tesla P100—a GPU with an architecture focused on accelerating AI and deep learning. Apparently, Nvidia spent more than \$2 billion on R&D to produce the new chip, which has a total of 15 billion transistors—around thrice as many as the company's previous chips. According to a press release, an artificial neural network powered by this chip can learn from incoming data twelve times faster than Nvidia's previous best chip.

To show that machine learning requires a special architecture, Google also developed its own processor called the tensor processing unit (TPU). On the other hand, Microsoft went with field-programmable gate arrays (FPGAs) for its AI needs. Not to be left out of the race, Intel acquired Altera—a company that sells FPGAs to Microsoft. Together with Nervana Systems, a company it acquired last year,

Deep learning—the most obvious trend in AI today

Deep learning is basically a nature-inspired technology that uses neural networks modelled after the human brain, to help a system learn by itself, just like a human child does. It is made of multi-layered deep neural networks that mimic the activity of the layers of neurons in the neocortex. Each layer tries to understand something more than the previous layer, thereby developing a deeper understanding of things. The resulting system is self-learning, which means that it is not restricted by what it has been taught to do. It can react according to the situation and even make decisions by itself.

Deep learning usually requires large memory banks and runs on huge servers powered by advanced graphics processing units. However, recent innovations have opened up the possibility of implementing deep learning at the network edge.

Find out more about deep learning: Deep Learning Makes Conventional Machine Learning Look Dumb (EY Jan. 2017)